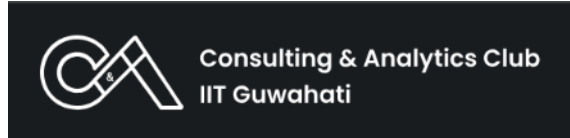


Time Series Analysis



What you'll learn:

- Time Data Visualisation in Python
- Autocorrelation
- Model selection and the role residuals play in it.
- Stationarity and how to test for its existence.
- Facebook's Prophet Library
- AR | MA | ARMA | ARIMA
- Deep learning (RNNs and LSTMs) for time series forecasting

WEEK 1	WHAT'S IN THERE	TASK 1	TASK 2	TASK 3
Day 1	Hey there, excited to start learning? Welcome to the fourth edition of our course on Time Series Analysis and Forecasting We'll start by exploring the fundamental concepts of time series forecasting, along with key statistical principles, to build a strong foundation for what's ahead.	What is TSA? ↗ Why is TSA important? ↗	Theoretical aspects of Time Series forecasting ↗	What are p-values? ↗
Day 2	Today, we'll delve into hypothesis testing, understand autocorrelation, and explore essential Python tools for time series analysis.	Hypothesis Testing - 1 ↗ Hypothesis Testing - 2 (till 4:37) ↗ Hypothesis Testing - 3 ↗	Understanding Autocorrelation and Partial Autocorrelation Functions ↗	Python basics for TSA (till module 6) ↗
Day 3	Now, let's explore the fundamentals of time series, key modeling techniques, and how to analyze residuals and prediction intervals.	Handbook for Handling Date Time ↗ Log Returns ↗	Residuals and Prediction Intervals ↗	Residuals Theory ↗
Day 4	Today, we'll dive into key time series concepts like white noise, log returns, and stationarity. We'll also explore the Augmented Dickey-Fuller (ADF) test to assess stationarity in time series data.	White Noise ↗	TS Basics ↗	ADF Test - 1 ↗ ADF Test - 2 ↗
Day 5	Before applying time series models we need to know data analysis and smoothing methods to get rid of noise.	Analysis of TS Data ↗	Smoothing Methods (upto 40 minutes) ↗	Notebook on Smoothing Methods

WEEK 2	WHAT'S IN THERE	TASK 1	TASK 2	TASK 3
Day 1	This week begins with an exploration of different data preprocessing and imputation techniques for handling missing data in time series.	Data Preprocessing ↗	Imputation Methods (Blog) ↗ Imputation Methods (Video) ↗	Imputation Notebook ↗
Day 2	Now we will learn time series modelling method ARMA & ARIMA. Here we combine autoregression and moving average method	ARMA ARIMA ↗	Notebook ↗	Video on the Notebook ↗
Day 3	Now, we use our knowledge of time series to model financial data.	Forecasting Future Sales Using ARIMA and SARIMAX ↗	Stock Market Price Trend Prediction Using Time Series Forecasting ↗	
Day 4	Now that we've covered the basics of time series, it's time to explore Facebook Prophet — an open-source algorithm designed for building powerful and accurate time series models.	Basics of Prophet 1 ↗ Basics of Prophet 2 ↗	Sample Forecasting using Prophet ↗	Diving deeper into Prophet ↗ Notebook ↗
Day 5	Let's learn how to forecast complex time series data using multivariate and multiple time series models, and understand event-time predictions through survival analysis.	Multivariate TS Forecasting ↗	Multiple Time series Modeling; ↗	Follow-along Video ↗

WEEK 3	WHAT'S IN THERE	TASK 1	TASK 2	TASK 3
Day 1	We will be using time series techniques for sales forecasting.	Kaggle notebook ↗	Sales & Demand Forecasting	
Day 2	Ever wondered how to model a time series in which variance is varying with time? In such a scenario ARCH/GARCH model is what one needs. Also, today we learn about survival analysis	Theoretical Aspects of ARCH/GARCH ↗ Volatility Forecasting (GARCH & ARCH) ↗	Introduction to Survival Analysis ↗	Notebook ↗
Day 3	Don't want to code a lot? Let's learn about AutoTS(Automatic Time Series), which is a software tool that automates machine learning (AutoML) to build and deploy time series forecasting models.	Reader [only up to Ensemble, Caveats and Advice are optional] ↗	AutoML TS ↗	
Day 4	Today, we will be learning about the Darts library which is available in python and has some great tools for making time series forecasting easier.	Darts Tutorial ↗	Notebook ↗	Anomaly Detection ↗
Day 5	Let's have a look at how neural networks and deep learning approach can help in time series analysis.	MLP	Gradient Descent ↗	Forward & Backward Propagation ↗

WEEK 4	WHAT'S IN THERE	TASK 1	TASK 2	TASK 3
Day 1	This notebook will take you through models such as LSTM and RNN to accomplish these tasks.	Recurrent Neural Networks ↗	Understanding LSTM ↗	Notebook ↗ Watch the video corresponding to the notebook (if required) ↗
Day 2	From where we left off, now let's continue to expand our knowledge of using deep learning model to multivariate time series forecasting	Theoretical aspects of Multivariate Time Series forecasting using DL [Refresher] ↗	End-to-End Multivariate TS Modelling using LSTM ↗	
Day 3	Sometimes time series data have a natural hierarchical structure and we use Hierarchical time series analysis to analyse and model such data	Forecasting Hierarchial TS ↗	Coding Hierarchical Time Series forecasting Part 1 ↗	Coding Hierarchical Time Series forecasting Part 2 ↗
Day 4	Let's explore how hybrid models are made for time series forecasting with their applications in real-world	Forecasting Competition ↗	N Beats ↗	
Day 5	Let's see how our time-series models are performing on unseen data using different validation methods. Also, now we know how to use ML in time series forecasting. Let's see how not to use ML in time series forecasting (Task 3)	Validation Methods - Video ↗	Validation Methods - notebook ↗	Article ↗

<https://www.caciitg.com/>
<https://www.caciitg.com/resources/tsa/#>
<https://chat.whatsapp.com/DotlRFTw7KzLfqkBZqeipw>
<https://youtu.be/GE3JOFWTWVM?si=trhZpf158PZ6opb0>
<https://www.youtube.com/watch?v=whml0WHuIK4>
<https://medium.com/@varun030403/time-series-forecasting-theoretical-aspects-part-1-9adf7b1e0ce3>
<https://www.youtube.com/watch?v=KS6KEWaoOOE>
<https://www.youtube.com/watch?v=XHPiEp-3yC0>
<https://www.youtube.com/watch?v=4mDemwLHzbo>
<https://youtu.be/JkLCPOZcc4A?si=JKEMw6SWNWruUgueb>
<https://medium.com/@kis.andras.nandor/understanding-autocorrelation-and-partial-autocorrelation-functions-acf-and-pacf-2998e7e1bcb5>
<https://www.kaggle.com/code/prashant111/complete-guide-on-time-series-analysis-in-python>
<https://jakevdp.github.io/PythonDataScienceHandbook/03.11-working-with-time-series.html>
<https://quantivity.wordpress.com/2011/02/21/why-log-returns/>
<https://youtu.be/MtJwFpqJBil?si=Guq4l4zOiYQlnQiz>
<https://www.statology.org/residuals/>
<https://otexts.com/fpp2/wn.html>
<https://www.kaggle.com/code/varunbhardwaj03/ts-basics>
<https://www.machinelearningplus.com/time-series/augmented-dickey-fuller-test/>
<https://www.numberanalytics.com/blog/10-key-facts-augmented-dickey-fuller-test>
<https://www.youtube.com/watch?v=3sH1kisAK9s&list=PL3N9eeOICrP5cK0QRQxeJd6GrQvhAtpBK&index=2>
https://www.youtube.com/watch?v=kAl67Sz92-s&list=PL98nY_tJQXZmT9ZB59T0lsx0ZzzLrYdX4&index=1
<https://www.kaggle.com/code/konradb/ts-1a-smoothing-methods/notebook>
<https://www.youtube.com/watch?v=jR0phoeXjrc>
<https://www.geeksforgeeks.org/how-to-deal-with-missing-values-in-a-timeseries-in-python/>
<https://www.youtube.com/watch?v=GEytNZVjZNU>
https://www.kaggle.com/code/iamtushara/timeseries-imputation-techniques-complete-guide?utm_source=chatgpt.com
<https://365datascience.com/tutorials/time-series-analysis-tutorials/arma-model/>
<https://towardsdatascience.com/understanding-arima-time-series-modeling-d99cd11be3f8/%22>
<https://www.kaggle.com/code/konradb/ts-2-linear-vision/notebook?scriptVersionId=106896888>
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<https://www.analyticsvidhya.com/blog/2020/11/stock-market-price-trend-prediction-using-time-series-forecasting/>
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<https://www.youtube.com/watch?v=D8CFPyi4ai4&list=PL3N9eeOICrP5cK0QRQxeJd6GrQvhAtpBK&index=11>
<https://www.kaggle.com/code/konradb/ts-1b-prophet/notebook>
<https://www.youtube.com/live/XZhPO043lqU>
<https://www.youtube.com/watch?v=UhxCsweuKE4&list=PL3N9eeOICrP5cK0QRQxeJd6GrQvhAtpBK&index=16>
https://www.youtube.com/watch?v=74rDhJexmTg&list=PL98nY_tJQXZmT9ZB59T0lsx0ZzzLrYdX4&index=3

<https://www.kaggle.com/code/konradb/ts-4-sales-and-demand-forecasting/notebook>
https://www.youtube.com/watch?v=RdH8zd07u2E&list=PL98nY_tJQXZmT9ZB59T0lsx0ZzzLrYdX4&index=4
<https://medium.com/@simomenaldo/volatility-modeling-a-deep-dive-into-the-arch-family-f361bafbb3f7>
https://colab.research.google.com/github/Nixtla/statsforecast/blob/main/nbs/docs/tutorials/GARCH_tutorial.ipynb
https://youtu.be/Wo9RNcHM_bs?si=-ehn9x6ebX25o3KM
<https://www.kaggle.com/code/gunesevitan/survival-analysis/>
<https://winedarksea.github.io/AutoTS/build/html/source/tutorial.html>
<https://medium.com/data-science-in-your-pocket/automl-for-time-series-forecasting-using-autots-with-example-735fb559443a>
<https://www.youtube.com/watch?v=Sx-ul-PypmU>
<https://www.kaggle.com/code/konradb/ts-5-automatic-for-the-people/notebook%22>
<https://www.youtube.com/watch?v=hkXPdkPfgoo>
<https://www.datacamp.com/tutorial/multilayer-perceptrons-in-machine-learning>
<https://youtu.be/sDv4f4s2SB8?si=eABQNsf8GkXSeB9I>
<https://www.youtube.com/watch?v=Tb23YtZ92AE>
<https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-recurrent-neural-networks>
<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>
<https://www.kaggle.com/code/konradb/ts-6-deep-learning-for-ts-rnn-and-friends?scriptVersionId=108938187>
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<https://towardsdatascience.com/multivariate-time-series-forecasting-with-deep-learning-3e7b3e2d2bcf>
<https://www.youtube.com/watch?v=4FmVlpcwl4k>
<https://otexts.com/fpp2/hierarchical.html>
<https://towardsdatascience.com/introduction-to-hierarchical-modeling-a5c7b2ebb1ca>
<https://towardsdatascience.com/introduction-to-hierarchical-time-series-forecasting-part-ii-e73dfd3de86b>
<https://www.uber.com/en-IN/blog/m4-forecasting-competition/>
<https://towardsdatascience.com/n-beats-time-series-forecasting-with-neural-basis-expansion-af09ea39f538>
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<https://www.kaggle.com/code/konradb/ts-10-validation-methods-for-time-series/>
<https://towardsdatascience.com/how-not-to-use-machine-learning-for-time-series-forecasting-avoiding-the-pitfalls-19f9d7adf424>
https://unstop.com/p/time-series-analysis-week-1-iit-guwahati-1673358?lb=Qh0loyl&utm_medium=Share&utm_source=caciitgoff&utm_campaign=Quizzes
<https://unstop.com/quiz/time-series-analysis-week-2-iit-guwahati-1676318>
https://unstop.com/o/y1F9Si3?lb=Qh0loyl&utm_medium=Share&utm_source=caciitgoff&utm_campaign=Quizzes
https://unstop.com/o/Op9i60y?lb=Qh0loyl&utm_medium=Share&utm_source=quizzes&utm_campaign=Caciitgoff
<https://www.iitg.ac.in/sa/caciitg>

Assessment



LINK	DEADLINE	INSTRUCTIONS
Week 1 Quiz 	11:59 PM on April 12, 2026	Use the same email id that you used for registration
Week 2 Quiz 	11:59 PM IST on April 19, 2026	
Week 3 Quiz 	11:59 PM on April 27, 2026	Use the same email id that you used for registration
Week 4 Quiz 	11:59 PM on May 4, 2026	Use the same email id that you used for registration



Online



Time Series Analysis: Week 1

Indian Institute of Technology (IIT), Guwahati



Team Size

Individual Participation



Registration Deadline

14 Apr 26, 11:59 PM IST

Quizzes & Treasure Hunt

Stages and Timelines

10
Apr

10 Apr 26, 05:00 PM IST → 14 Apr 26, 11:59 PM IST

Time Series Analysis 2026: Week 1 Quiz

On Unstop

This will be a simple quiz of 12 questions! You will see the Play button here, once the round is live

Results

All that you need to know about Time Series Analysis: Week 1

This is a weekly assessment, part of the Time Series Analysis 2026 course offered by C&A, IIT Guwahati. It tests concepts covered in the course material up till now. The quiz emphasizes logical thinking, problem-solving, and applying learned concepts.

Guidelines:

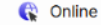
Eligibility: Open to all registered students of the Time Series Analysis 2026 course.

Quiz: 12 MCQs based on the Week 1 course material.

Duration: 45 minutes

Maximum Marks: 36

Important: Participants must use the same Unstop account they used for registering for the Time Series Analysis course.



Online



Time Series Analysis: Week 2

Indian Institute of Technology (IIT), Guwahati



Team Size

Individual Participation



Registration Deadline

20 Apr 26, 11:59 PM IST

Quizzes & Treasure Hunt

Stages and Timelines

17
Apr

17 Apr 26, 05:00 PM IST → 20 Apr 26, 11:59 PM IST

Time Series Analysis 2026: Week 2 Quiz

On Unstop

This will be a simple quiz of 9 questions! You will see the Play button here, once the round is live.

Results

All that you need to know about Time Series Analysis: Week 2

This is a weekly assessment, part of the Time Series Analysis 2026 course offered by C&A, IIT Guwahati. It tests concepts covered in the course material up till now. The quiz emphasizes logical thinking, problem-solving, and applying learned concepts.

Guidelines:

- **Eligibility:** Open to all registered students of the Time Series Analysis 2026 course.

- **Quiz:** 9 MCQs based on the Week 2 course material.

- **Duration:** 45 minutes

- **Maximum Marks:** 27

- **Important:** Participants must use the same Unstop account they used for registering for the Time Series Analysis course.

Time Series Analysis: Week 3

Indian Institute of Technology (IIT), Guwahati



-  **Team Size**
Individual Participation
-  **Registration Deadline**
28 Apr 26, 11:59 PM IST

Quizzes & Treasure Hunt

Stages and Timelines

24 Apr 24 Apr 26, 05:00 PM IST → 28 Apr 26, 11:59 PM IST

Time Series Analysis 2026: Week 3 Quiz

On 

This will be a simple quiz of 10 questions! You will see the Play button here, once the round is live.

Results

All that you need to know about Time Series Analysis: Week 3

This is a weekly assessment, part of the Time Series Analysis 2026 course offered by C&A, IIT Guwahati. It tests concepts covered in the course material up till now. The quiz emphasizes logical thinking, problem-solving, and applying learned concepts.

Guidelines:

- **Eligibility:** Open to all registered students of the Time Series Analysis 2026 course.
- **Quiz:** 10 MCQs based on the Week 3 course material: <https://caciitg.com/resources/tsa/>
- **Duration:** 45 minutes
- **Maximum Marks:** 30
- **Important:** Participants must use the same Unstop account they used for registering for the Time Series Analysis course.

Time Series Analysis: Week 4

Indian Institute of Technology (IIT), Guwahati



-  **Team Size**
Individual Participation
-  **Registration Deadline**
05 May 26, 11:59 PM IST

Quizzes & Treasure Hunt

Stages and Timelines

1 May 01 May 26, 05:00 PM IST → 05 May 26, 11:59 PM IST

Time Series Analysis 2026: Week 4 Quiz

On 

This will be a simple quiz of 20 questions! You will see the Play button here, once the round is live.

Results

All that you need to know about Time Series Analysis: Week 4

This is a weekly assessment, part of the Time Series Analysis 2026 course offered by C&A, IIT Guwahati. It tests concepts covered in the course material up till now. The quiz emphasizes logical thinking, problem-solving, and applying learned concepts.

Guidelines:

- **Eligibility:** Open to all registered students of the Time Series Analysis 2026 course.
- **Quiz:** 20 MCQs based on the Week 4 course material.
- **Duration:** 50 minutes
- **Maximum Marks:** 60
- **Important:** Participants must use the same Unstop account they used for registering for the Time Series Analysis course.



Time Series Analysis: Week 1

📍 Online

📅 Updated On: Apr 25, 2026

🌐 Official website ↗

Results

Your Result

546

Prashant Ranjan

Rank	Score
546	30

Overall Result

Rank	Name / Organization	Score	
1	Niharika Sikka Manipal University (MU), Jaipur	36.000	🔗
2	HANVITH SAI ALLA VNR VJIET	36.000	🔗
3	Mohd Azad Bansal Institute of Engineering and Technology, Lucknow	36.000	🔗

1 - 3 / 3

Jump To 1



Time Series Analysis: Week 2

Online

Updated On: Apr 20, 2026

Results

Your Result

319

Prashant Ranjan

Rank

319

Score

27

Top Gainers

TA

2nd Rank

27.000
Score

TANYA GARG

SD College of Engineering and Technology,
Muzaffarnagar

YA

1st Rank

27.000
Score

Yadavalli Sandeep

Indian Institute of Management (IIM),
Calcutta

SU

3rd Rank

27.000
Score

Suraj Kumar Oraon

Atma Ram Sanatan Dharma College
(ARSD), University of Delhi (DU), New Delhi



Time Series Analysis: Week 3

Online

Updated On: Apr 28, 2026

Official website ↗

Results

Your Result



Prashant Ranjan

Score

26.67

Overall Result

Rank	Name / Organization	Score	
1	Tejas Bramhe Indian Institute of Technology (IIT), Kharagpur	30.000	
2	Janhvi Dhanraj Hingankar Shri Ramdeobaba College of Engineering and Management (RCOEM), Nagpur	30.000	
3	Shubham Kumar Saharsa College of Engineering(SCE),Saharsa	30.000	

1 - 3 / 3

Jump To 1



Time Series Analysis: Week 4

📍 Online

📅 Updated On: May 4, 2026

Results

Your Result

444

Prashant Ranjan

Rank

444

Score

54

🏆 Top Gainers

YU

2nd Rank

60.000
Score

Yuvraj Gupta

Indian Institute of Technology (IIT),
Guwahati

KU



1st Rank

60.000
Score

Kushank Gupta

Ramjas College, University of Delhi

HA

3rd Rank

60.000
Score

Harshit Pareek Pareek

Indian Institute of Technology (IIT),
Guwahati

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Ask Gemini

Summarize, analyze, and get up to speed with files in this folder.

Type

People

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Source

Name	Owner	Date modified	File size	Sort
Capstone Project TSA 2026.pdf	caciitg.storage	May 8	3.4 MB	

31



Consulting & Analytics Club
IIT Guwahati



Capstone Project 2026

TIME SERIES ANALYSIS 2026

IMPORTANT LINKS

CAPSTONE SUBMISSION LINK: [CLICK HERE](#)

VIRTUAL TRADING ON STOCKGRO (DETAILS INSIDE)

1. CAPSTONE REGISTRATION FORM: [CLICK HERE](#)

2. TRADING PLATFORM LINK: [CLICK HERE](#)



Consulting & Analytics Club
IIT Guwahati



CAPSTONE PROBLEM STATEMENT

Data-Driven Stock Analysis using Time Series Models on StockGro

OBJECTIVE

In this capstone project, you will apply time series forecasting and analysis techniques to select stocks and allocate capital within a virtual ₹10,00,000 portfolio on the StockGro platform. You will build predictive models (ARIMA, Prophet, LSTM, etc.) on historical NSE stock data, use those models to make informed investment decisions, execute those decisions on the StockGro virtual trading platform, and finally compare your model's predictions against the actual market outcomes observed on StockGro. The entire workflow — from data analysis to live virtual trading to final reporting — constitutes your submission.

PLATFORM BACKGROUND

StockGro is a virtual stock market simulator that uses live NSE/BSE data. It allows users to build portfolios, allocate funds, execute trades, and monitor real-time performance — all without any real financial risk. You will use StockGro as the execution environment to test whether your data-driven forecasts translate into actual portfolio gains.



DATASET SPECIFICATION

- Source: Yahoo Finance via the `yfinance` Python library
- Period: 1st January 2021 to 31st December 2025
- Interval: Daily (1d)
- Train/Test Split: Use a minimum of the last 6 months of this period (i.e., July 2025 – December 2025) as your test/validation set. Everything before that is your training data.

Example code to fetch data:

```
1 import yfinance as yf
2 data = yf.download("RELIANCE.NS", start="2021-01-01", end="2025-12-31", interval="1d")
```

STEP-BY-STEP PROJECT TASKS

Follow the tasks below in order. Each task builds on the previous one.

TASK 1: STOCK UNIVERSE SELECTION

What you must do:

- Choose stocks from diverse sectors (e.g., Banking, IT, Pharma, FMCG, Auto) to ensure portfolio diversification.
- Justify each stock selection using at least one of the following data-driven approaches:
 - Rolling standard deviation to identify stocks with interesting volatility profiles
 - Seasonal decomposition to detect stocks with strong trends
 - Sector-based rationale (e.g., selecting from a high-momentum sector)

Deliverable: A written justification for your chosen stock universe.



TASK 2: DATA PREPROCESSING

Before modeling, prepare your data so it is clean and suitable for time series analysis.

What you must do:

- Handle missing values using forward fill, backward fill, or removal
- Test each stock's price series for stationarity using the Augmented Dickey-Fuller (ADF) test; apply differencing if the series is non-stationary
- Normalize or scale data if you plan to use ML-based models like LSTM
- Split the data into training (before July 2025) and testing (July 2025 – December 2025) sets

Deliverable: Clean, preprocessed time series datasets for each selected stock, ready for modeling.

TASK 3: TIME SERIES FORECASTING

Build forecasting models to predict stock prices for the next 2 trading days beyond your dataset (i.e., into early January 2026, or equivalently, the 2 days of the StockGro trading window).

What you must do:

- Implement at least one (preferably two or more) of the following models:
 - ARIMA / SARIMA
 - Exponential Smoothing (SES / Holt-Winters)
 - Facebook Prophet
 - LSTM / GRU
 - CNN-based time series model
 - Any transformer-based or pretrained model
- For each model:
 - Train on the training set
 - Validate on the test set (last 6 months)
 - Forecast prices for the next 5 trading days
 - Evaluate using MAPE, RMSE, and/or directional accuracy (did the model correctly predict whether the price goes up or down?)
- Tune ARIMA using AIC/BIC criteria and validate residuals
- Optionally, combine multiple model predictions into an ensemble for better accuracy

Deliverable: Forecasted prices for each stock over the next 2 trading days, along with model evaluation metrics.

TASK 4: VOLATILITY AND TREND ANALYSIS

Beyond just price forecasts, you must understand each stock's risk profile and trend behavior.

What you must do:

Volatility Estimation:

- Compute log returns for each stock
- Analyze fluctuations in log returns to estimate short-term risk
- Optionally use GARCH(1,1) or rolling standard deviation to model time-varying volatility and forecast near-term uncertainty

Trend and Pattern Analysis:

- Apply seasonal decomposition (e.g., STL decomposition) to break each stock's price series into trend, seasonality, and residual/noise components
- Evaluate the strength and direction of the trend to understand whether a stock is in an upward, downward, or sideways phase

Deliverable: Volatility estimates and trend assessments for each stock in your universe, which will directly feed into your portfolio allocation strategy in the next task.



TASK 5: PORTFOLIO CONSTRUCTION AND CAPITAL ALLOCATION

Using the outputs of Tasks 3 and 4, decide how to allocate your ₹10,00,000 virtual capital across your selected stocks. You must use at least two of the following strategies:

Strategy A — Forecast-Guided Allocation:

- Rank stocks by their predicted 5-day return from your forecasting models
- Allocate more capital to stocks with higher predicted gains and higher forecast confidence
- Optionally adjust weights based on model uncertainty or prediction intervals

Strategy B — Volatility-Aware Sizing:

- Use predicted volatility (from GARCH or rolling std dev) to assign weights inversely proportional to volatility: $w_i = (1/\hat{\sigma}_i) / \sum_j (1/\hat{\sigma}_j)$
- Stocks expected to be more volatile receive a smaller allocation; more stable stocks receive a larger share
- Normalize all weights to sum to 1

Strategy C — Correlation-Based Diversification:

- Compute a correlation matrix across your selected stocks using recent return windows
- Prefer combinations of stocks that are lowly correlated with each other to reduce portfolio-level risk
- Avoid concentrating capital in highly correlated stocks

Strategy D — Sector Momentum Rotation:

- Group your selected stocks by sector
- Assess which sectors are showing consistent upward momentum
- Allocate more capital to high-momentum sectors, and within each sector, use one of the above strategies to finalize individual stock weights

Deliverable: A final portfolio allocation table showing: stock name, allocated amount (in ₹), weight (%), and the reasoning/strategy behind the allocation.



TASK 6: MODEL COMPARISON

Compare all the forecasting models you built in Task 3.

What you must do:

- Tabulate MAPE, RMSE, and directional accuracy for each model across all stocks
- Comment on each model's strengths and weaknesses (e.g., ARIMA is interpretable but struggles with non-linearity; LSTM captures complex patterns but requires more data and tuning)
- Discuss whether an ensemble approach improved results
- State clearly which model(s) you ultimately relied on for your portfolio decisions and why

Deliverable: A comparative model evaluation table and a written discussion.



TASK 7: VIRTUAL TRADING ON STOCKGRO

This is where you execute your data-driven strategy in the real (virtual) market.

STEP-BY-STEP INSTRUCTIONS:

1. Register for capstone ([Registration Link](#)) using the provided form to gain access to the Live Event.
2. Log in at : <https://stockgro.onelink.me/vNON/2ijikjck>
 - Log in with your Registered mobile Number
 - Go to Trackers in the app → find Event : “Portfolio – Time Series Analysis 2026”
 - Make sure your StockGro app is updated, otherwise the Event Portfolio will not be visible
3. Open the Event Portfolio and start building your portfolio
4. Deploy your ₹10,00,000 by purchasing the stocks in the exact proportions determined in Task 5
5. The trading window is open Monday to Friday, **11 May 2025 – 15 May 2025 (market hours: 9:35 AM – 3:25 PM each day)**. Pick any two consecutive trading days within this window as your personal execution and observation window. Execute all your trades before the market close of your chosen Day 1, and record your portfolio performance at the market close of your chosen Day 2.

VALID DAY PAIRS YOU CAN CHOOSE FROM:

- DAY 1: MON 11 MAY → DAY 2: TUE 12 MAY
- DAY 1: TUE 12 MAY → DAY 2: WED 13 MAY
- DAY 1: WED 13 MAY → DAY 2: THU 14 MAY
- DAY 1: THU 14 MAY → DAY 2: FRI 15 MAY

To confirm enrollment: if “Portfolio – Time Series Analysis 2026” appears in your Trackers, your registration is successful.

TUTORIAL VIDEO: [CLICK HERE](#)



TASK 8: PERFORMANCE TRACKING AND COMPARISON AGAINST YOUR PREDICTIONS

This is the most critical task — comparing what your models predicted against what actually happened in the market. Throughout this task, “Day 1” and “Day 2” refer to whichever consecutive day pair you selected in Task 7.

STEP 1 – RECORD ACTUAL OUTCOMES:

- At the market close of your Day 2 (3:25 PM), note the actual closing prices of all your selected stocks from the StockGro app
- Record your portfolio's total return, individual stock returns, and any volatility observed across the two days

STEP 2 – COMPARE PREDICTIONS VS REALITY:

- For each stock, place your model's forecasted price for Day 1 and Day 2 alongside the actual prices observed on StockGro
- Compute the prediction error (MAPE or absolute % error) for this live two-day window
- Check directional accuracy: did your model correctly predict whether each stock went up or down?

STEP 3 – PORTFOLIO PERFORMANCE ANALYSIS:

- Report your total portfolio return (%) over your chosen two-day window
- Identify which stocks contributed positively and which dragged performance
- Assess whether your chosen allocation strategy from Task 5 held up under actual market conditions

STEP 4 – REFLECT:

- Did your model predictions align with actual market movement?
- Did higher-confidence stocks actually perform better?
- Was your volatility estimation accurate? Did volatile stocks behave as expected?
- What would you change in your model, preprocessing, or allocation strategy if you were to redo this?

Deliverable: A structured comparison table (predicted vs actual prices and returns) and a written reflection.

FINAL SUBMISSION REQUIREMENTS :

Your submission must include all of the following:

1. Python Notebook(s) (.ipynb):
 - Full pipeline: data fetching → preprocessing → modeling → forecasting → portfolio construction
 - All code must be runnable and clearly commented
 - Reference sample notebook provided:
[stock_price_prediction_using_yfinance\[1\].ipynb](#)

2. REPORT (MAXIMUM 10 PAGES):

Your report must cover the following sections in order:

- Methodology and models used
- Stock selection rationale (Task 1)
- Data preprocessing steps (Task 2)
- Forecast results with confidence intervals (Task 3)
- Volatility and trend analysis findings (Task 4)
- Portfolio composition and allocation rationale (Task 5)
- Model comparison and accuracy evaluation (Task 6)
- StockGro execution summary — trades placed, capital deployed (Task 7)
- Predicted vs actual outcomes — this is the core of your report (Task 8)
- Reflections: what worked, what didn't, and what you would improve

3. OPTIONAL BONUS — VISUAL DASHBOARD:

Build an interactive or static dashboard that includes:

- Forecast plots: actual vs predicted prices for each stock
- Portfolio allocation chart (pie or bar chart by stock and sector)
- Correlation heatmap (if using diversification strategy)
- Trend and volatility graphs supporting your strategy rationale

SUBMISSION LINK: [CLICK HERE](#)

ALL THE BEST
~TEAM C&A



Capstone Project Submission - Time Series Analysis 2026

•Welcome to the Capstone Project Submission Portal for the *Time Series Analysis Course*, conducted in collaboration with **StockGro – India's leading stock market simulation platform**.

This project is designed to help you apply time series forecasting concepts to real-world stock market scenarios using models such as ARIMA, Prophet, LSTM, GARCH, and other relevant techniques. Participants are expected to develop forecasting strategies, build a virtual portfolio, and evaluate their performance during the live trading simulation event.

Problem Statement Link : https://drive.google.com/drive/folders/1J7KXFJ0BRrUrTANF9MQGH78H_S-EBA0u?usp=sharing

Important Timeline 8th May – Project Release

(Problem statement, guidelines, and submission details shared)

•**10th May EOD** - Registration Closes

•**11th May – 15th May** – Live Trading Event

(Apply your forecasting model and trading strategy in real-time simulation)

•**17th May (EOD)** – Final Submission Deadline

•**Important Notes** Start early and test your models thoroughly before the trading event begins.

•Ensure that all submitted work is original.

•Late submissions may not be considered.

•Follow all guidelines mentioned in the project statement carefully.

We look forward to seeing your innovative forecasting strategies and analytical approaches.

All the best! 🍀

Consulting and Analytics Club, IIT Guwahati

1.Name

2.Registered E-mail ID

3.Contact Number

4.Link to your Final Submission Code File And Report (please ensure the drive folder has read access for everyone with link)

5.Bonus Task : Visual Dashboard (Github or Live Link)